Machine Learning 02450 - Assignment 1

Group 510
Simon Fogh Kristiansen - s194135
Christian Damén Schultz-Nielsen - s240369
Benjamin Chambaudet - s240414

February 2024

Student ID	Dataset description	Data attribute desc.	Data Visualization	Exam Questions
s194135	30%	30%	40%	33.33%
s240369	30%	40%	30%	33.33%
s240414	40%	30%	30%	33.33%

Table 1: Contributions of the three collaborators for this report.

1 Dataset description

The chosen dataset depicts an analysis of the chemical composition of different wine from the same region in Italy. The data comes from three different cultivars and should therefore provide data from different sites. The data includes the quantities of 13 different attributes of the wines from the 3 different wine cultivars. The wine data was found using a data repository also linked to in the exercises[4]. The chosen dataset is to be found at: `https://archive.ics.uci.edu/dataset/109/wine`.

Next, 2 former uses of our data is briefly described:

- Previously the data has been used for comparing different classifiers in [2]. If one reads the article, it compares 8 different methods e.g. LDA (Linear Discriminant Analysis), RDA (Regularized Discriminant Analysis), and 1NN(k-Nearest Neighbour with k equal to 1). It is concluded that only RDA achieved 100% on classifying data from the Wine Dataset used in this group.

- In the article [1] *The data was used as an illustrator of superior performance for a new appreciation function using Regularized Discriminant Analysis. The dataset was also used to compare Error counting (EC) and The geometric appreciation function (GAF).*

We want to describe the chemical compounds in relation with regression and use classification for determining the origin of the wine from the 3 vineyards. In this work, we hope to learn both which attributes are good for this and which aren't. Perhaps attributes follow each other, perhaps they are uncorrelated and this is to be investigated. The classification is also expected not to be straightforward, so classification models and understanding how to choose the optimal one is also in our learning objective. An example of what has been spoken about is, if the protein concentration and proline could be used for classification as depicted on Figure 1.

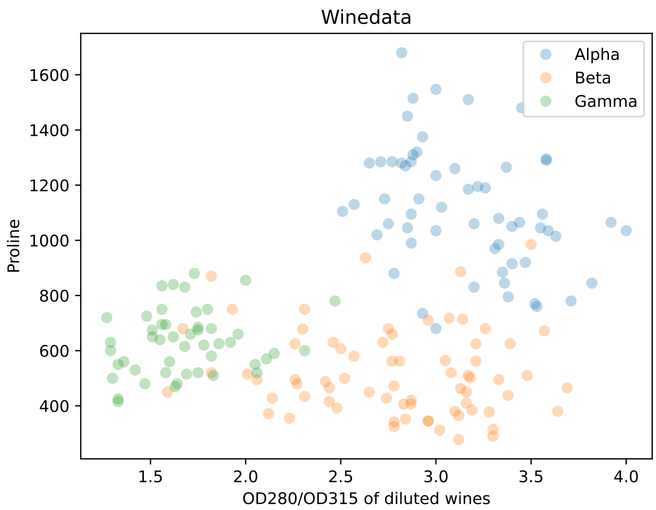


Figure 1: The attributes protein concentration and proline plotted against each other with the different wine origins being different colors and the Greek letters of Alpha, Beta and Gamma.

Another attribute that we would like to predict from other attributes is the ash which is a measure of the residue which is left after evaporating the liquid in the wine and afterwards igniting the physical compound at 500 °C to 550 °C. The leftover is the ash.

We expect the wines to be somewhat closely related in chemical composition, since it is wine after all and not other types of beverages with much more variance. Therefore, a PCA analysis must be relevant for exploring trends in the data which in a lower dimension can be analysed and worked with.

For classifying which three of the vineyards the wines come from, there is no need for data transformation as would have been the case if classification was done on a continuous variable, where a limit should be found to determine, if the data was lower or higher than some value. The three vineyards is deducted to be a discrete and nominal variable.

2 Data attribute description

As said before, our dataset counts 13 attributes which are listed in the table below with short descriptions: All attributes are **continuous** and are **ratio** type with exception of the wine origin *Cultivars* which is **discrete** and **nominal**. It seems there are no missing values in the dataset and to check if the data have issues with outliers, we looked at the maximum/minimum value for each attribute. By loading the dataset into python, the dataset was checked for corrupt and missing value by testing/inspection, if each value in the dataset was a number or a NoneType element. From the test it was seen that there was no missing or corrupt values in the dataset.

Attribute	Description
1) Cultivars	Which vineyard the wine comes from
2) Alcohol	Percentage of alcohol in wine.
3) Malic acid	Grams of malic acid per liter.
4) Ash	Ash content in grams per liter.
5) Alkalinity of ash	Alkalinity of ash in mEq/l.
6) Magnesium	Magnesium content in mg/dm^3.
7) Total phenols	Total phenolic content in mg/dm^3.
8) Flavanoids	Flavanoid content in mg/dm^3.
9) Nonflavanoid phenols	Nonflavanoid phenolic content in mg/dm^3.
10) Proanthocyanins	Proanthocyanin content in mg/dm^3.
11) Color intensity	Color intensity.
12) Hue	Hue.
13) OD280/OD315 of diluted wines	OD280/OD315 ratio of diluted wines.
14) Proline	Proline content in mg/dm^3.

Table 2: Attributes of Wine

The following Table 3 the basic statistical summary is given. This is done through the attributes' mean, standard deviation, median, and range. This is chosen as the basic statistic summary, as this shows a lot of information about how the datapoints are distributed and in what range they are.

Attribute	Mean	Standard dev.	Median	Range
Alcohol	13.00	0.81	13.05	3.80
Malic acid	2.34	1.12	1.87	5.06
Ash	2.37	0.27	2.36	1.87
Alcalinity of ash	19.49	3.34	19.50	19.40
Magnesium	99.74	14.28	98.00	92.00
Total phenols	2.30	0.63	2.36	2.90
Flavanoids	2.03	1.00	2.14	4.74
Nonflavanoid phenols	0.36	0.12	0.34	0.53
Proanthocyanins	1.59	0.57	1.56	3.17
Color intensity	5.06	2.32	4.69	11.72
Hue	0.96	0.23	0.96	1.23
OD280/OD315 of diluted wines	2.61	0.71	2.78	2.73
Proline	746.89	314.91	673.50	1402.00

Table 3: Basic summary statistics of the attributes.

3 Dataset visualization

3.1 Attribute Analysis

In order to check if there is any problem with outliers in the wine data, a box plot with the different attributes has been made. It can be seen in figure 2. From figure 2 it is seen that the number of outliers are minimum and the magnitude of the outliers are not to far from the rest of the data.

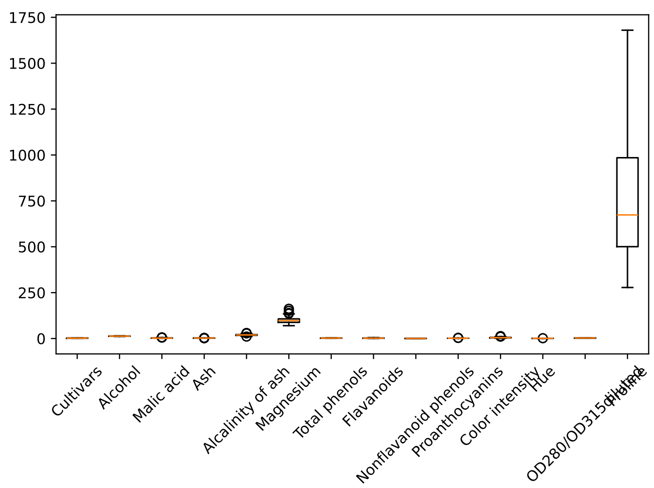


Figure 2: Box plot of the wine data

The data is normalized too, just to make sure that the Proline doesn't hide some serious outliers in the other attributes. This is shown in figure 3. Her it is once again seen that, the outliers are no problem.

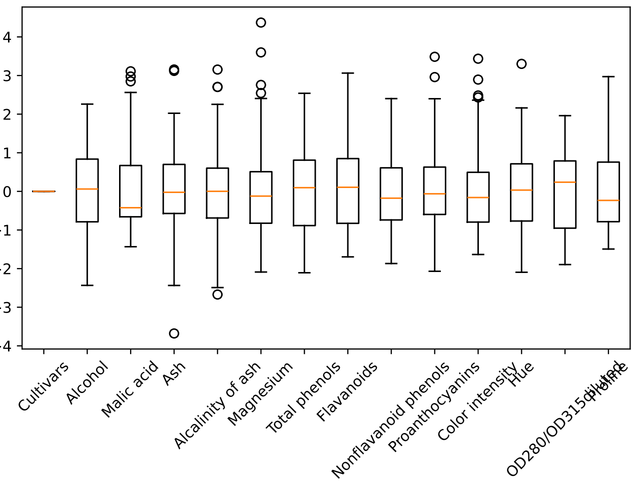


Figure 3: Box plot of the wine data

It is also investigated, if the attributes appear normally distributed, and this is done with histograms in figure 4. Here it is seen that some of the attributes are close to normally distributed, while others are not normally distributed.

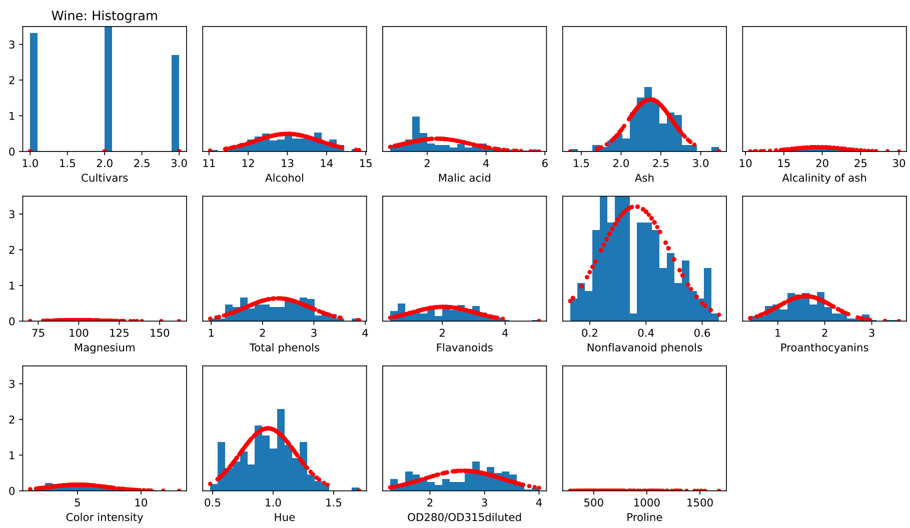


Figure 4: Histogram of the wine data

Now the data is investigated more directly, and this is done by creating scatter-plots between all the attributes. This is done here in Figure 5 Which shows the correlation between all 14 attributes.

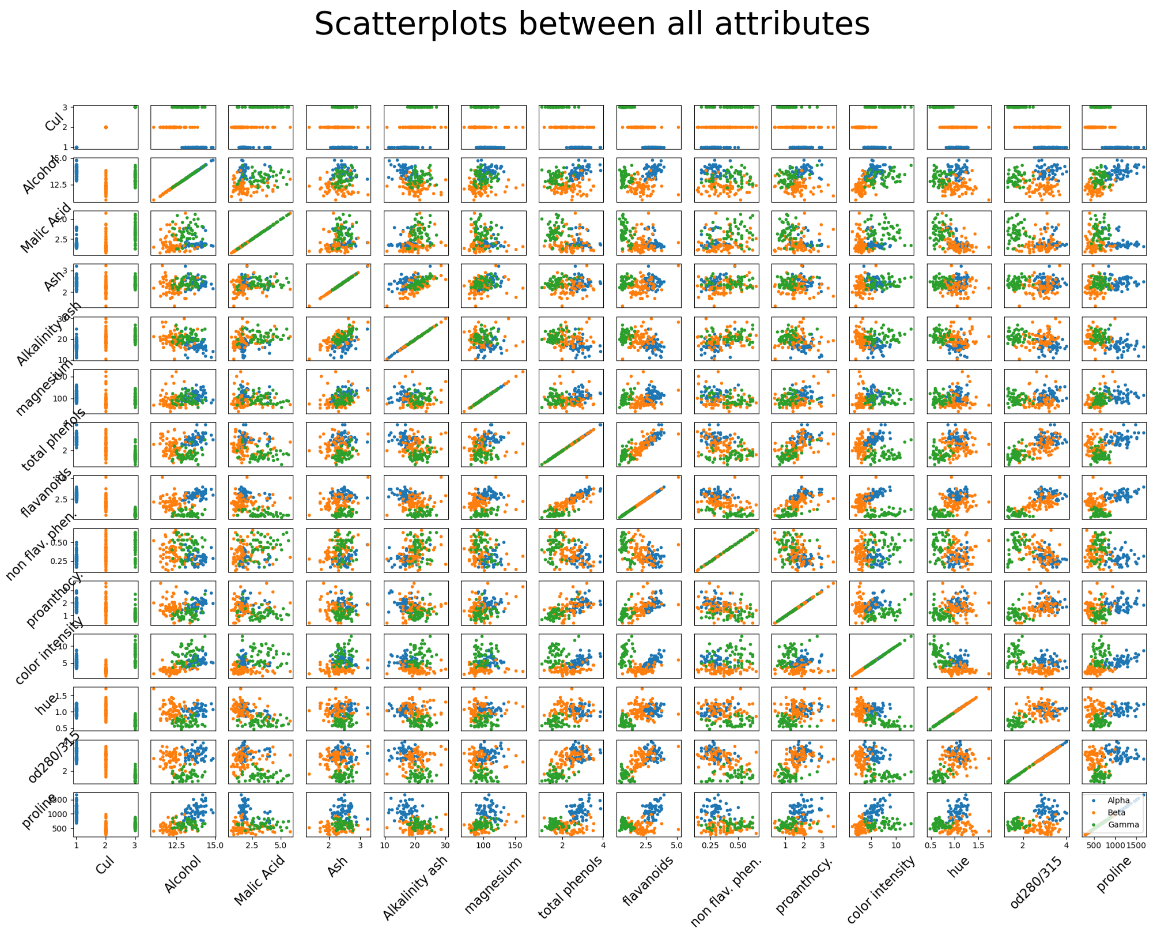


Figure 5: Scatterplots between all attributes. The correlation on the diagonal of course has a correlation of 1 since these are the variances. There is also symmetry, so the focus can be on one side of the diagonal, when searching the data.

Since there are a lot of attributes a 14 x 14 scatterplot for evaluating the correlation between them would be overwhelming. Therefore, we searched for another visualization method and decided to do a heatmap. This is used to answer, if the attributes are correlated. The correlation coefficient is used, so the data is normalised and the dependency between attributes is between -1 and 1. This is done here in Figure 6:

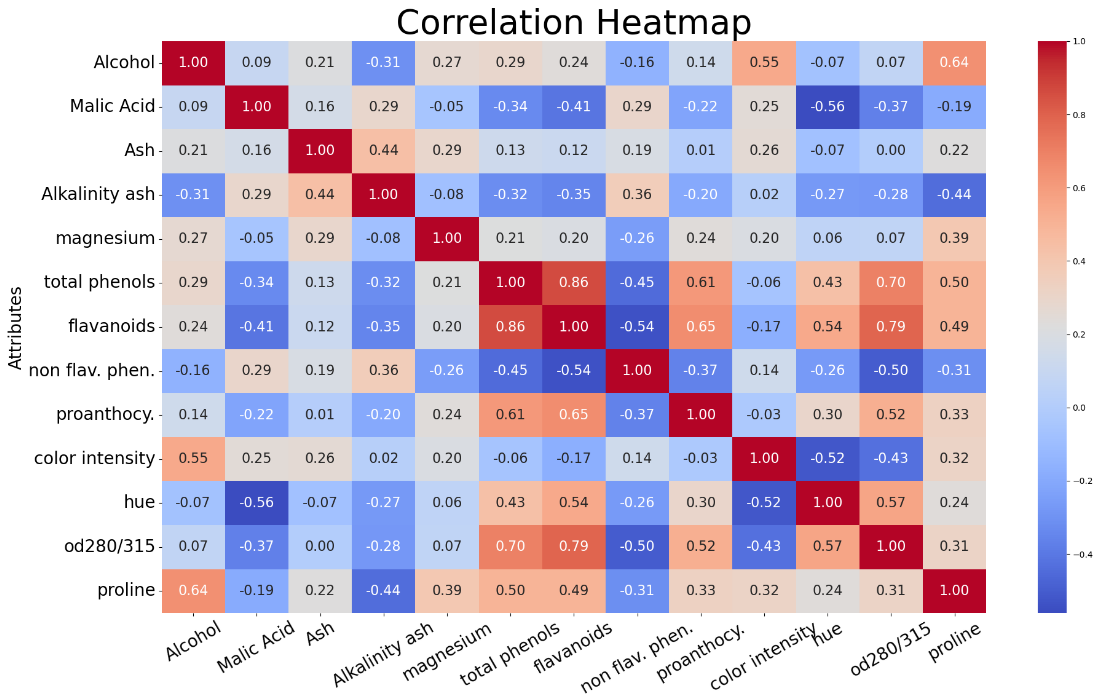


Figure 6: Heatmap of the correlation between attributes in the wine dataset.

Now the attributes are compared by plotting each attribute against all the other. It can here be seen that the covariance with highest correlation is flavanoids and total phenols. The least correlated attributes are Ash and Od280/315 and these are plotted in Figure 7.

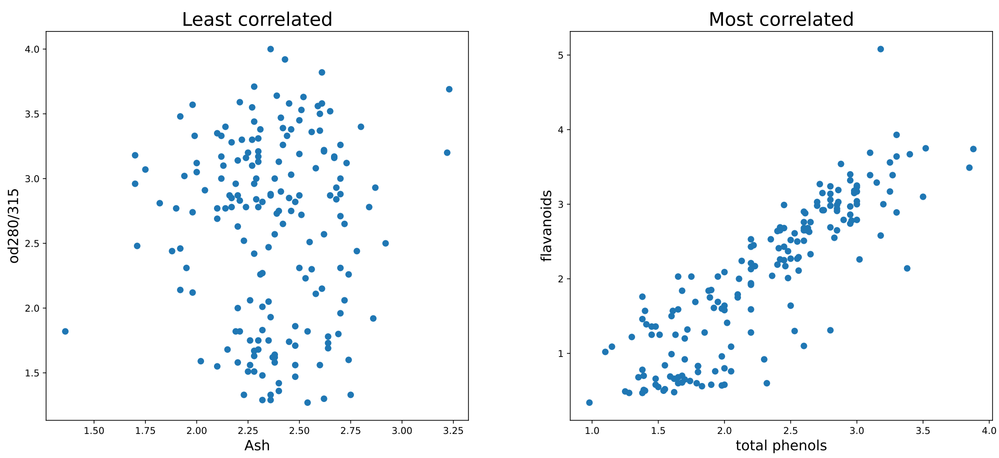


Figure 7: The most uncorrelated attributes are **OD280/315 vs. Ash** which has a correlation of 0.00 and the most correlated attributes are **flavanoids vs.total phenols** which has correlation of 0.86.

So there is clearly some attributes in the data where regression can be done upon. Also, with a look on the scatterplot Figure 5 and the Figure 1 there are some attributes where a classification task can be worked with. **Proline vs. OD280/315** was already found just be searching manually, and with the scatterplot, the **Proline vs. Hue** and **Alcohol vs. flavanoids** also seems like a good choice for the classification.

3.2 PCA Analysis

In this subsection the data is analysed with Principal Component Analysis.

3.2.1 Standardisation and Singular Value Decomposition

Firstly, the data needs to be standardised by subtracting each attribute with its own mean and dividing by its own standard deviation. This removes the dataweight in high variance data which would dominate the PCA analysis as learned in *Exercise 2.1.6*. In the following we denote $\boldsymbol{X}_i$ as the i'th vector (attribute) in the matrix of attributes. Therefore, the normalisation can be written as:

$$\tilde{\boldsymbol{X}}_i = \frac{\boldsymbol{X}_i - \mu_i}{\sigma_i} \tag{1}$$

And with the normalised data, the Singular Value Decomposition can be applied:

$$\tilde{\boldsymbol{X}} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T \tag{2}$$

Where Σ contains the singular values which are to be used in describing the variation in the components. V is the orthonormal basis of the lower dimension subspace of N principal components. The subspace is denoted as in [3]:

$$\boldsymbol{B} = \tilde{\boldsymbol{X}}\boldsymbol{V}_n \tag{3}$$

Where n is the number of principal components.

3.2.2 Variance explained with PCA components

The variance which is described in the different PCA components can be calculated with:

$$ExplainedVariance = \frac{\sigma_i^2}{\sum_{i=1}^{M} \sigma_i^2} \tag{4}$$

This allows us to determine, both how descriptive the single PCA components are, but also determining how many components are needed for some percentage of the variance being described. This can be done by adding the variance of the first M components:

$$ExplainedVariance = \frac{\sum_{i=1}^{K} \sigma_i^2}{\sum_{i=1}^{M} \sigma_i^2} \tag{5}$$

This is done, and Figure 8 shows that 8 components are needed to describe 90% of the variance in the data.

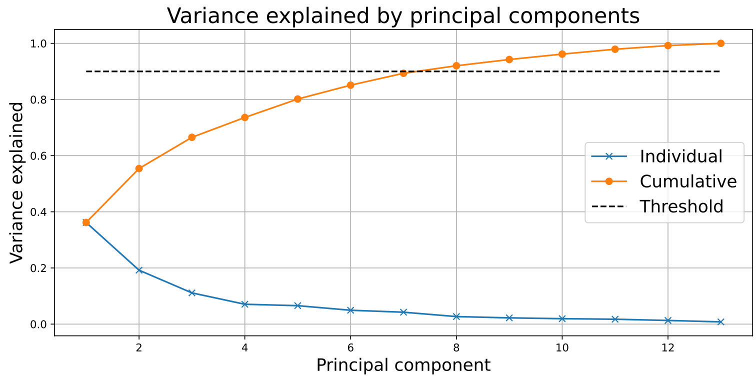


Figure 8: Explained variance plotted both individually and cumulative. Threshold of 90% showing that 8 principal components are needed to describe 90% of the data.

3.2.3 Principal directions of PCA components

The directions of PCA components show how well an attribute is described by the different PCA components. This is done and the result seen in Figure 9

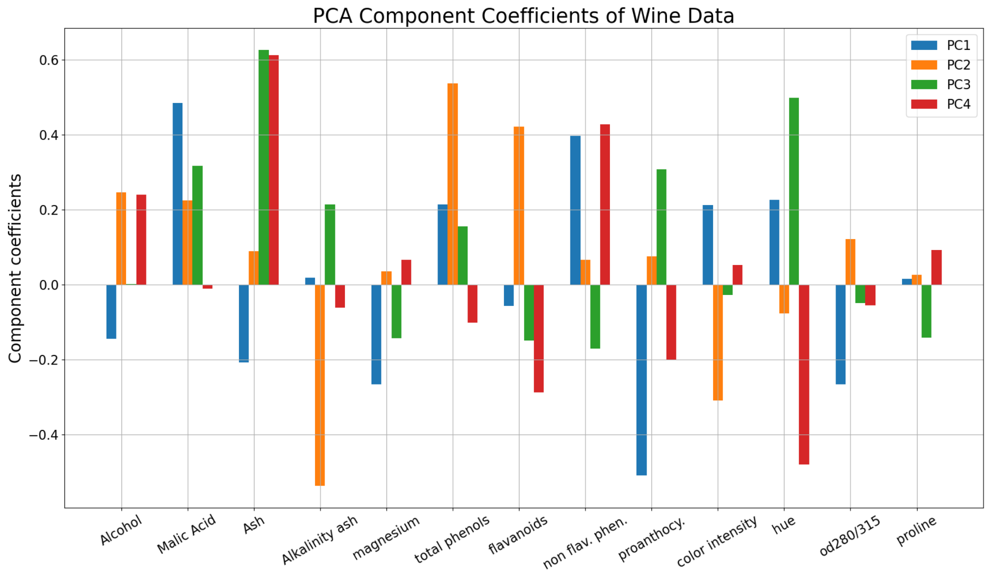
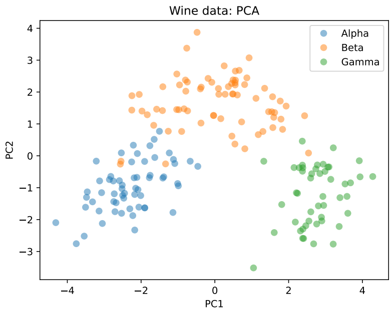


Figure 9: Plot of the vector decomposition of the 4 PCA's for each attribute. The plot descries how much of the variance can be explained with each PCA for the different attributes. The first four components are depicted, to not over complicate the plot.

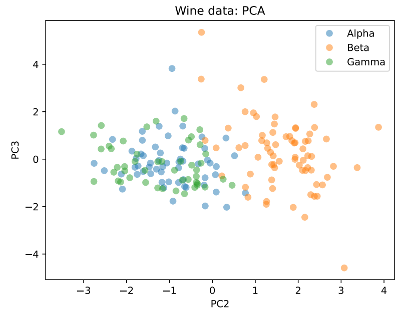
Here it can be seen that Ash is best described with PC3 and PC4 and worst with PC2. The directions plotted can therefore show, where an attribute is described in the lower dimensional vector space.

3.2.4 Data projected onto PCA components

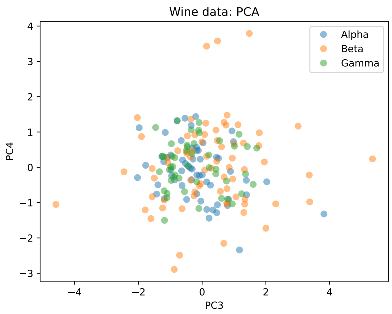
In Figure 10 four plots of the different principle components are plotted against each other. It is seen that in plot Figure 10(c) has the most similarity, since the points in the PCA plot lies in one big cluster and Figure 10(a) has the least similarity because it have the most separation of the points. Note, that PC1 to PC4 are not the only PCA components, just the highlighted here.



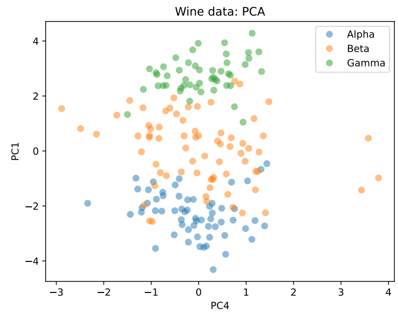
(a) PC1 vs PC2



(b) PC2 vs PC3



(c) PC3 vs PC4



(d) PC4 vs PC1

Figure 10: The 4 different PCA plotted against each other. From the plots it is seen that the most distinctions is in plot a) and d).

4 Discussion

What we have learned: There are not missing, corrupt or NoneType in the dataset. We also learned that some of the attributes are must more dominate compared to others, and this is seen in the boxplot in Figure 2 where Proline by far is the most dominant of the attributes. The dataset also have no issues with outliers which is seen in Figure 3. Our aim is to determine, from which cultivar the wine is coming from. From Figure 10(a) and Figure 10(b) it is seen that there is a reasonable separation in the data to determine from which cultivar the wine is coming from. So by selecting the right attributes from the data, classification is feasible in categorizing the data and determining from which cultivar the wine belongs. Furthermore, regression is a goal for the work, since there are some correlated attributes which can be used to predict other attributes.

5 Exam Problems

5.1 Question 1. Spring 2019 question 1:

From the 4 answers it is seen that **the correct answer is D**. Where x1 is an interval, which is the interval between each measurements. x6 is a ratio of how many traffic lights is broken. x7 is also a ratio, of how many run over accidents are in the dataset interval. And y is a ordinal of how much traffic is on the road.

5.2 Question 2. Spring 2019 question 2:

The simplest way to do this to compute all the answers and see which one is correct. This is done by writing the 2 vectors into python and using linalg.norm to compute the p-norm distance. The results gives

$$d_{p=\infty}(x_{14}, x_{18}) = 7.00$$

So from the calculating from python it can be seen that **the correct answers is A.**

5.3 Question 3. Spring 2019 question 3:

For each proposition we calculate the specified fractions of explained variance for the i'th component and sum them accordingly:

$$ExplainedVariance = \frac{\sum_{i=1}^{K} \sigma_i^2}{\sum_{i=1}^{M} \sigma_i^2} \tag{6}$$

- The variance explained by the first four principal components is equal to 0.867

We conclude that right answer is the answer A.

5.4 Question 5. Spring 2019 question 14:

The Jaccard similarity is given the the total number of attributes K, the number of attributes which are similar between different vectors f_{11} and the number of attributes which are not similar between the vectors f_{00}. Since there are two similar words (no stemming, so identical) plus stop words are not removed, then 2 words are similar with 6 being different. So:

$$J(s_1, s_2) = \frac{f_{11}}{K - f_{00}} = \frac{2}{20000 - 6} = 0.00010003 \approx 0.000100 \tag{7}$$

Meaning **C is the correct answer**

5.5 Question 6. Spring 2019 question 27:

Since the table shows the conditional probabilities, we already know the probabilities of all outcomes, given that $y = 2$. Therefore, we just have to add the probabilities of the outcomes, where $\hat{x}_2 = 0$. There are two such outcomes, where $\hat{x}_7$ varies, so **B is the correct answer**:

$$p(\hat{x}_2 = 0|y = 2) = p(\hat{x}_2 = 0, \hat{x}_7 = 0|y = 2) + p(\hat{x}_2 = 0, \hat{x}_7 = 1|y = 2) = 0.81 + 0.03 = 0.84 \tag{8}$$

References

[1] D. Coomans S. Aeberhard and O. de Vel. *THE CLASSIFICATION PERFORMANCE OF RDA*. Tech. rep. Dept. of Computer Science, Dept. of Mathematics, and Statistics, James Cook University of North Queensland, 1992.

[2] D. Coomans S. Aeberhard and O. de Vel. *Comparative analysis of statistical pattern recognition methods in high dimensional settings*. Tech. rep. Dept. of Computer Science, Dept. of Mathematics, and Statistics, James Cook University of North Queensland, 1994.

[3] Mikkel N. Schmidt Tue Herlau and Morten Mørup. *Introduction to Machine Learning and Data Mining*. Technical University of Denmark, 2023.

[4] *UCI Machine Learning Repository*. `https://archive.ics.uci.edu`. Accessed: 21-02-2024.